Positive octahedra and primitive fillings for trees of crossing circles

Abstract

An ordered tree of crossing circles has two ideal octahedra for each physical quarter-turn in its tangle construction. We give explicit holomorphic shapes for these cells and follow the original peripheral curves through their gluing. If a circle has degree d_v and is filled to crossing power n_v, the condition $\sqrt{d_v^2 + n_v^2} \geq 630$ at every circle produces a positive cone deformation from the augmented exterior to the smooth filled exterior. The bound is independent of the number of circles and allows zero powers at sufficiently large degrees. We identify the absolute metric volume with a holomorphic critical action, bound the core lengths and volume loss, and give a more flexible criterion with a different parameter radius at each vertex. The argument concerns the original marked link complement and does not require a quantum asymptotic.

1 The marked diagrams and the geometric statement

A crossing circle is an auxiliary component surrounding two strands; its spanning disk is punctured twice by those strands. Removing full twists while retaining the circle gives an augmented link. Filling the circle along its disk-framed slope restores the original twists. The marking of that slope matters: a full turn and a trivial endpoint permutation are different operations.

We use two-string tangles with four ordered endpoints, vertical stacking, a physical quarter-turn T, and cylindrical closure C. The underlying finite tree $\mathcal{T}$ has at least one edge. Each vertex carries its own crossing-circle tangle P_v; there are no bare product junctions. In a rooted description, a nonroot tangle is

$$W_v = \left(\prod_{w \in I_v^-} T(W_w) \right) P_v \left(\prod_{w \in I_v^+} T(W_w) \right). \tag{1.1}$$

The two lists have their specified order and record the cut at the parent edge in the original cyclic order. One may choose a terminal root and close its tangle with the adjacent turned branch. All rotations of child balls, endpoint frames, and disk orientations are retained. In particular a double turn is the physical half-turn, which reverses the order of a stack; it is not erased by a scalar identity.

Let e be the number of tree edges, and $d_v \geq 1$ the degree of vertex v. Choose $\epsilon_v \in \{-1, 0, 1\}$ and an integer p_v, and put

$$n_v = \epsilon_v - 2p_v. \tag{1.2}$$

The augmented diagram retains the crossing R^{ϵ_v} in the original local frame. Its exterior is denoted Y^ϵ. The original meridian μ_v and disk longitude λ_v form a primitive basis on the crossing-circle torus. Our sign convention for R is that filling $\mu_v + p_v\lambda_v$ restores R^{n_v}. Section 3.2 specifies the two odd meridian words, so this convention includes both signs of a retained half twist. To see the underlying surgery, cut along the twice-punctured disk

and rotate one side through p_v full turns before removing the auxiliary circle. The slope becomes the meridian of the added solid torus, and the two strands acquire $-2p_v$ crossings in this disk frame. This local operation takes place in a ball in S^3; it does not change the ambient manifold. Write $Y(\mathbf{p})$ for the result of filling only these auxiliary tori. The main components are the strands left after filling. Their number is determined by the literal endpoint matching, not by the abstract tree. If that matching gives one component, write it as $K_{\mathbf{n}}$; then $Y(\mathbf{p}) = S^3 \setminus K_{\mathbf{n}}$.

Let v_{oct} be the volume of a regular ideal octahedron. For complex t, u near zero define the holomorphic function ψ by

$$\psi(0,0) = \frac{v_{\mathrm{oct}}}{\pi}, \qquad \psi_t(t,u) = -\frac{1}{\pi}\operatorname{arsinh}(\tan(t/2)\cos(u/2)), \qquad \psi_u(t,u) = \psi_t(u,t). \tag{1.3}$$

All branches continue from zero. The equality of the cross derivatives, proved in Section 4, makes this a definition rather than an extra existence assumption. Put

$$\Psi(\boldsymbol{\tau}) = \sum_{vw \in E(\mathcal{T})} \psi(\tau_v, \tau_w), \qquad L_v = \sqrt{d_v^2 + n_v^2}, \qquad \delta = \frac{1}{39000}.$$

Theorem 1.1. *For the marked diagrams above, suppose $L_v \geq 630$ for every vertex. There is a unique solution in $|\tau_v| \leq 10^{-2}$ of*

$$\Psi_{\tau_v}(\boldsymbol{\tau}) + \frac{\mathrm{i}n_v}{2\pi}\tau_v = \mathrm{i}c. \tag{1.4}$$

It is holomorphic in c on a neighborhood of the closed unit disk, vanishes at $c = 0$, and for real $0 < c \leq 1$ satisfies

$$|\tau_v(c)| \leq \frac{2\pi c}{L_v - \delta d_v}, \qquad \operatorname{Im}\tau_v(c) < 0. \tag{1.5}$$

The original augmented exterior admits a positive ideal-cell metric with auxiliary logarithmic periods

$$U_v = -2\pi\Psi_{\tau_v} - \mathrm{i}\epsilon_v\tau_v, \qquad V_v = 2\mathrm{i}\tau_v, \qquad U_v + p_vV_v = -2\pi\mathrm{i}c. \tag{1.6}$$

These are the full periods of the primitive curves, continued from completeness. Completing the auxiliary ends gives cone angle $2\pi c$ and core length $\ell_v(c) = -2\operatorname{Im}\tau_v(c)$. At $c = 1$ the completion is smooth, complete, and of finite volume on $Y(\mathbf{p})$; all main cusps remain complete.

Define the stationary value, at the solution of (1.4),

$$\mathcal{C}(c) = \Psi(\boldsymbol{\tau}(c)) + \frac{\mathrm{i}}{4\pi}\sum_v n_v\tau_v(c)^2 - \mathrm{i}c\sum_v \tau_v(c). \tag{1.7}$$

The absolute volume of the completed cone metric is

$$\mathrm{Vol}_c = 2\pi\operatorname{Re}\mathcal{C}(c), \qquad \mathrm{Vol}_0 = 2e\,v_{\mathrm{oct}}. \tag{1.8}$$

In particular, if the main component is a knot, the right side at $c = 1$ is $\mathrm{Vol}(S^3 \setminus K_{\mathbf{n}})$.

The theorem is a sufficient geometric criterion for each fixed finite diagram. There is no bound on its size, no restriction on the signs of n_v, and no lower bound on $|n_v|$ alone. A vertex with $n_v = 0$ is allowed when $d_v \geq 630$. The conclusion does not assert a quantum invariant limit: that requires an additional finite-source identity and a noncancellation estimate. Nor is this a theorem about every strict Conway presentation; the presence of a crossing circle at each product junction is part of the construction.

The geometry of augmented links, their crossing-disk decompositions, and the role of regular octahedra are described by Purcell [2]. Gluing along thrice-punctured spheres has a

substantial history; see Morgan–Ransom–Spyropoulos–Trapp–Ziegler [3] for a recent account of belted decompositions. We prove the ordered cut and its peripheral markings directly here. The only specialized companion input is the local peripheral-volume differential of [1, Section 7, equation (7.3)]; its hypotheses and conventions are stated in Section 6. In particular we do not invoke that paper's independently long-filling theorem.

2 The projection cut and its ordered octahedra

Opening the projection regions and crossing disks gives two projection copies. We call disk-side faces shaded and projection faces white. An ideal vertex records either a main strand segment or an auxiliary semicircle. The following explicit construction identifies the cut with the original diagram before any shape is assigned.

For a turned leaf $h_v = T(P_v)$ label its half-segments

$$(v_0, v_1, v_2, v_3) = (\text{lower left}, \text{lower right}, \text{upper right}, \text{upper left}).$$

A lower factor F followed by an upper factor G joins F_2 to G_1 and F_3 to G_0, leaving the tuple (F_0, F_1, G_2, G_3). Closure joins positions 0 to 3 and 1 to 2. These are empty joining strips in the projection plane. The inverse turn $P_v = T^{-1}(h_v)$ has exposed tuple (v_3, v_0, v_1, v_2).

Let O_v denote the auxiliary ideal vertex. Its two shaded triangles are

$$(O_v, v_1, v_2), \qquad (O_v, v_3, v_0). \tag{2.1}$$

A half-segment in this notation means the ideal vertex containing its matched pair. The cyclic order around O_v is (v_0, v_1, v_2, v_3). At a main vertex the corner sectors concatenate in the order of the planar joining strip. Following the complementary sectors gives all white face cycles. This determines the oriented boundary ribbon graph, including more information than just the numbers of faces or components.

The cut is reversible. An auxiliary semicircle is boundary-parallel in its projection half-ball. Drilling it creates a handle, and opening its recorded half disk cancels that handle. The disjoint leaf boxes allow these operations simultaneously. Regluing the white faces by their projection labels and the shaded faces by their original disk maps restores the original exterior. A retained half twist changes the disk regluing, not this open cut.

Proposition 2.1. *Each projection copy is a ball obtained by attaching e octahedra along $e-1$ triangular faces. Every edge of $\mathcal{T}$ contributes one octahedron in each copy. At vertex v the incident blocks give d_v consecutive disk-side pages, of which exactly one cyclic seam carries the original retained crossing R^{ϵ_v}. All main segments, disk curves, and white face cycles agree with the original marked exterior.*

Proof. Consider insertion of a terminal child h_w immediately after P_v. If x_1, x_2 are the outside partners of v_1, v_2, the product rule is

before	after	
$v_1 - x_1$	$v_1 - w_1,$	$w_2 - x_1$
$v_2 - x_2$	$v_2 - w_0,$	$w_3 - x_2.$

(2.2)

Write $C = (v_1, x_1)$ and $B = (v_2, x_2)$ for the old vertices. After insertion these are (w_2, x_1) and (w_3, x_2); the new vertices are $D = (v_1, w_1)$ and $A = (v_2, w_0)$. Attach an octahedron with apices O_v, O_w, equator (A, B, C, D), and shaded faces

$$(O_v, B, C), \quad (O_v, D, A), \quad (O_w, D, C), \quad (O_w, B, A). \tag{2.3}$$

Its first face is glued to the old face (O_v, C, B) with opposite orientation. The remaining three faces are exactly the new faces specified by (2.1). The other shaded face at v is unchanged.

The four complementary faces are white triangles. At B, C, O_v remove the glued shaded sectors and concatenate the remaining sectors. Their cyclic orders are precisely those of the two strip replacements in (2.2); the new vertices A, D are the two new identity strips. Consequently the white boundary walks also agree. Doing the same in the other projection copy adds one pair of octahedra and one original circle. It introduces no auxiliary circle at the triangular interface.

Insertion before P_v uses the other face and the equally explicit table

$$v_3 - x_3 \longmapsto (v_3 - w_3, w_0 - x_3), \qquad v_0 - x_0 \longmapsto (v_0 - w_2, w_1 - x_0). \tag{2.4}$$

This follows from the same product rule in the inverse-turn tuple. Both constructions are reversible by deleting the terminal octahedron.

For the initial edge the two-leaf endpoint matching gives exactly (2.3) in each projection copy. Choose a terminal root of the prescribed tree. Starting at its unique edge, insert the right children in reverse final order immediately after their parent circle, and the left children in final order immediately before it. Subsequent insertions take place inside the new child balls. This constructs the specified lists in (1.1); it does not commute existing factors. A turn of a child carries the entire substitution, including the projection copies and all four endpoint labels.

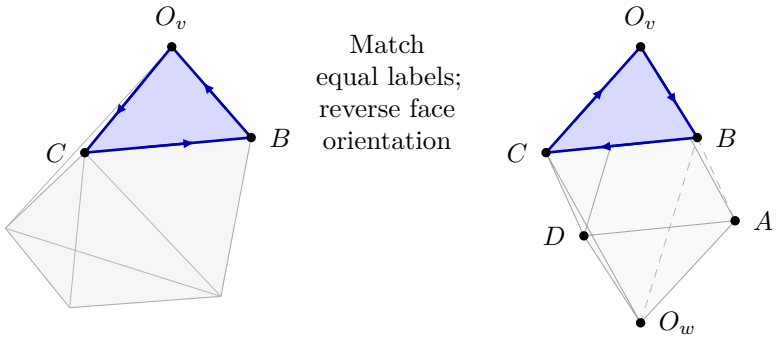
If the chosen root is displayed as $C(h_0 T(E))$, its collar can be drawn as $C(P_0 T^2(E))$. The half-turn T^2 carries a stack $L_1 \cdots L_k$ to $T^2(L_k) \cdots T^2(L_1)$ and commutes with the physical quarter-turn of each ball. Retaining this rotation gives the same initial edge and induction. Thus this collar change does not erase a half-turn or permute an open product.

Each step attaches a ball along a triangular disk, so the projection copy remains a ball. Its shaded face count is $4e - 2(e-1) = 2(e+1)$, agreeing with the two faces per circle already identified individually. The equal oriented ribbon graphs give a label-preserving homeomorphism of boundary spheres. Extending over the balls and applying their recorded face pairings identifies the quotient with the original marked exterior.

An upper interface triangle and its lower mate form a two-triangle thrice-punctured sphere. Its ends are the original auxiliary circle and its two main punctures; compactification gives a twice-punctured disk page. The insertion order puts these pages in the original cyclic order. There are $d_v - 1$ new seams and one distinguished original disk seam at v. Reglue the latter by its original half twist only once. Reversing a disk frame transports its two projection copies, labels, and both peripheral paths together. This completes the marked claim. □

2.1 The primitive collar

On the auxiliary torus the d_v blocks supply d_v annular sectors. Every page boundary is the same primitive disk longitude λ_v. The original meridian crosses these sectors once, in their cyclic order. A terminal insertion subdivides a passage into two consecutive passages; it adds neither winding nor a second longitude. Equivalently, the new triangular interface appears as a transition and its inverse in the concatenated face word. They cancel, leaving one primitive transverse passage and the distinguished original disk map. Locally the two rectangles of a crossing-circle collar identify opposite sides once. Their side curves are μ_v, λ_v, hence their intersection is one. For an odd seam one endpoint identification is shifted, but the two rectangles and their once-crossing paths are retained. Section 3.2 computes those two odd words rather than replacing the shift by an integral half longitude.



Existing projection ball
shaded face (O_v, C, B)

New octahedron
for edge vw
shaded face (O_v, B, C)

The glued triangle becomes internal. The three other shaded faces of the octahedron retain the circle v and introduce the circle w.

Figure 1: One terminal insertion in one projection copy. The old shaded triangle (O_v, C, B) is identified with (O_v, B, C) in the new octahedron. The other three shaded faces are those in (2.3); the white sectors follow the two strand-strip replacements in (2.2). The same operation in the other projection copy gives the two-octahedron block of edge vw.

3 A holomorphic two-octahedron block

The two-leaf closure is $C(h_0T(h_1))$. Normalize a leaf quadrilateral to $(0, \infty, 1, R)$ and its auxiliary apex to Z. In the other projection copy the apex is Z/E, where

$$Z = \frac{1+E}{1+H}, \qquad R = \frac{H(1+E)^2}{E(1+H)^2}, \qquad Z^2 - (1+E)Z + ER = 0. \tag{3.1}$$

Here E, H are free complex parameters near $\mathrm{i}, 1$. The two copy coordinates are holomorphic, not complex conjugates. The normalized turn and closing maps are

$$T(z) = \frac{R_1 - 1}{z - 1}, \qquad C(z) = \frac{z - R}{z - 1}, \qquad R_1 = 1 - R. \tag{3.2}$$

They follow by prescribing the images of the three labeled vertices of the turned and closing faces. Both maps are required to place the second leaf.

For longitude parameters t, u, put

$$\begin{gathered} y = \tan(t/2), \quad z = \tan(u/2), \quad k = (1 + y^2 + z^2)^{-1/2}, \quad k(0,0) = 1, \\ E = \frac{1 + \mathrm{i}k}{1 - \mathrm{i}k}, \quad H_0 = \frac{1 - ky}{1 + ky}, \quad H_1 = \frac{1 - kz}{1 + kz}, \\ R = \frac{1 + z^2}{2 + y^2 + z^2}, \quad R_1 = 1 - R, \qquad Z_j = \frac{1+E}{1+H_j}. \end{gathered} \tag{3.3}$$

Substitution gives (3.1) for the first leaf, and its R_1 version for the second. Moreover

$$Q_j = \frac{E - Z_j}{Z_j - 1} = \frac{1 + \mathrm{i}\tan(\tau_j/2)}{1 - \mathrm{i}\tan(\tau_j/2)} = e^{\mathrm{i}\tau_j}, \qquad (\tau_0, \tau_1) = (t, u). \tag{3.4}$$

Thus these formulas prescribe the longitude multipliers exactly. One can also solve the matching directly: the two normalized base ratios are

$$\kappa_j = \frac{k^2(1 + \tan^2(\tau_j/2))}{1 - k^2\tan^2(\tau_j/2)}.$$

Their product is one precisely on the branch $(k^2+1)((1+y^2+z^2)k^2-1)=0$ through $k=1$. There $\kappa_0=(1+y^2)/(1+z^2)$ and the two ordered bases are complementary. This verifies the gluing rather than imposing a stationary quantum equation.

The octahedra share the equator $q=(0,\infty,1,R)$ and have apices

$$\begin{array}{c|cc} & A & B \\ \hline + & Z_0 & RZ_1/(Z_1-R_1) \\ - & Z_0/E & RZ_1/(Z_1-ER_1). \end{array} \tag{3.5}$$

The second column follows by composing the two maps in (3.2). At zero the rows are respectively $((1+\mathrm{i})/2,(1-\mathrm{i})/2)$ and their reversal.

Lemma 3.1. *For $|t|,|u|\le 10^{-2}$ each block is triangulated by eight positive ideal tetrahedra, whose shapes satisfy $|w-\mathrm{i}|<0.4$. Their face maps and internal edge returns are holomorphic throughout this bidisk.*

Proof. We use the standard orientation of upper half-space and the positive ordered model $(0,\infty,1,w)$, with cross ratio

$$w(a_0,a_1,a_2,a_3)=\frac{(a_3-a_0)(a_2-a_1)}{(a_3-a_1)(a_2-a_0)}. \tag{3.6}$$

For the plus copy take ordered tetrahedra (B_+,A_+,q_j,q_{j+1}), j modulo four; for the minus copy take (B_-,A_-,q_{j+1},q_j). The numerical shape in either case is the edge cross ratio

$$[a,b,c,d]=\frac{(a-c)(b-d)}{(a-d)(b-c)} \tag{3.7}$$

on the tuple with its first two entries reversed. All eight complete shapes are i. This distinction between a tetrahedron's orientation and the edge-ratio convention will also be used in the seam table. The diagonal between the two apices divides the original two-pyramid union into these four tetrahedra in each copy.

Elementary sine and cosine estimates give $|y|,|z|<0.005001$, hence

$$|y^2+z^2|<0.000051,\quad |k-1|<0.000026,\quad |E-\mathrm{i}|<0.00006,$$
$$|H_j-1|<0.0102,\qquad |R-\tfrac12|<0.000026.$$

Equation (3.3) gives $|Z_j-(1+\mathrm{i})/2|<0.004$. Each denominator in the last column of (3.5) has complete modulus $1/2$ and changes by less than 0.0041. Every finite apex therefore moves by less than 0.02. Every complete difference between distinct finite vertices occurring in a shape has modulus at least $1/2$. Each difference factor changes by relative size less than 0.08, so

$$|w-\mathrm{i}|\le\frac{4(0.08)+2(0.08)^2}{(1-0.08)^2}<0.4.$$

Factors involving infinity only shorten this estimate. Thus the vertices stay distinct and the tetrahedra stay positive. The product of the four shapes around the apex diagonal telescopes; the total angle is 2π on the branch through the regular octahedron. For complex t,u the equatorial quadrilateral need not lie in one plane. Its retained triangular boundary and tetrahedral subdivision are the geometric cells used here. □

3.1 The full disk-side boundary maps

The four white transitions of one leaf are

$$W_0 = \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}, \qquad W_1 = \begin{pmatrix} 1 & E-1 \\ 0 & E \end{pmatrix},$$
$$W_2 = \begin{pmatrix} E-R & R(1-E) \\ E-1 & 1-ER \end{pmatrix}, \quad W_3 = \begin{pmatrix} R & 0 \\ 1-E & ER \end{pmatrix}. \tag{3.8}$$

Matrices act projectively. Each is the unique map of its three labeled face vertices. For example $W_0(Z) = W_2(Z) = Z/E$ by (3.1). Across a shaded triangle, the white transition transports the triangle from the other projection copy. Put $Q = (E-Z)/(Z-1)$. The incoming triangle has cyclic vertices $(Z, \infty, 1)$; the outgoing one has $(Z, R, 0)$. Their doubled two-triangle disk-side boundaries have edge ratios

incoming edge	(Z,∞)	$(\infty,1)$	$(1,Z)$
ratio	$-Q^{-1}$	$-Q$	$-Q^{-1}$
outgoing edge	(Z,R)	$(R,0)$	$(0,Z)$
ratio	$-Q$	$-Q^{-1}$	$-Q$.

(3.9)

Here the ratio of an oriented edge (a,b) with adjacent third vertices c,d means (3.7). The first entry is $(1-Z)/(E-Z) = -Q^{-1}$. For the incoming base the opposite apex is $W_1^{-1}(Z/E) = Z - E + 1$; for the outgoing base it is $W_3^{-1}(Z/E) = ER/(2E-Z)$. These substitutions give the other base entries; the remaining entries follow from W_0, W_2 in the same way.

The complete complex ratios, including their shears, depend only on $Q = e^{\mathrm{i}\tau_v}$, not on the parameter at the other end of the edge. Outgoing and incoming boundaries therefore have inverse ratios under their original puncture matching. Normalize one triangle by its three ordered vertices. The ratio table then fixes the other triangle, so it identifies the whole boundary map. The main puncture products are one, and the auxiliary puncture product is $Q^{\pm 2}$ according to boundary orientation. The original longitude will have multiplier Q^2.

3.2 Disk words, signs, and logarithmic increments

The pure shaded map and the two cross-copy maps are

$$S = \begin{pmatrix} 0 & -ER \\ 1 & -1-E \end{pmatrix}, \quad A = \begin{pmatrix} R & -R \\ 1 & ER-1-E \end{pmatrix}, \quad B = \begin{pmatrix} R/E-1-1/E & R \\ -1 & R \end{pmatrix}. \tag{3.10}$$

The pure map sends $(Z, \infty, 1)$ to $(Z, 0, R)$ within one copy. The odd maps send respectively

$$A : (Z, \infty, 1) \mapsto (Z/E, R, 0), \qquad B : (Z, R, 0) \mapsto (Z/E, \infty, 1).$$

The reversal of the two punctures is the literal half-twist disk identification. Put $\bar{S} = S|_{E \mapsto E^{-1}}$. Direct multiplication, using $ER = Z(1+E-Z)$, gives the pure returns

$$W_0 S = \bar{S} W_0, \qquad W_2 S = \bar{S} W_2, \qquad W_3 S = \bar{S} W_1 \tag{3.11}$$

projectively, and the odd returns

$$W_0^{-1} B W_2^{-1} A = I, \quad W_2^{-1} B W_0^{-1} A = I, \quad W_1^{-1} B W_3^{-1} A = I \tag{3.12}$$

projectively. The respective scalar representatives in (3.12) are $-R/E, -R/E, (R-1)/E$ times the identity. These equations check complete edge maps, not just multipliers. Transport

to a common ordered pants chart by (3.9) gives the same specified odd seam for incident blocks with different far-end parameters. No arbitrary permutation of pants punctures is used.

For clarity, send the auxiliary flag Z to infinity. If a matrix M fixes Z, its affine multiplier there is $(M_{21}Z+M_{22})^2/\det M$, the reciprocal of its derivative at Z. Set $\rho = Z/(1+E-Z) = H^{-1}$. The two-rectangle paths of Section 2.1 are the following words:

curve	word	multiplier at infinity
λ	$W_2^{-1}W_0$	Q^2
μ $(\epsilon=0)$	S^{-1}	ρ
μ $(\epsilon=-1)$	$A^{-1}W_0$	ρQ
μ $(\epsilon=+1)$	$W_0^{-1}B$	ρ/Q.

(3.13)

All words fix Z, and the displayed multiplier formula verifies each entry algebraically. To read their signs topologically, start on the incoming disk rectangle, follow the meridian through its first side, and return across the white side. The negative half twist uses the inverse A passage followed by W_0; the positive half twist uses B followed by W_0^{-1}. The longitude runs across W_0 and returns across W_2. The paths cross once, as before insertion. This is also an explicit convention for the positive crossing R in (1.2). Reversing a physical disk frame transports these words and its signed crossing together.

The logarithm of the pure local multiplier is

$$\log\rho = -\log H_0 = 2\,\mathrm{artanh}(ky) = 2\,\mathrm{arsinh}(\tan(t/2)\cos(u/2)) = -2\pi\psi_t(t,u). \tag{3.14}$$

The middle equality uses $ky = y/\sqrt{1+y^2+z^2}$ and the branches at zero. Thus (1.3) has a direct peripheral meaning, independent of a source action. Concatenating the d_v page increments cancels changes of intermediate pants frames. The distinguished disk word adds exactly $-\epsilon_v \log Q_v$, giving the first two equations of (1.6). These are initially chosen logarithms of the actual multipliers. Section 5.1 proves that they are the full log-link periods along the entire path.

4 A uniform solution of the filling equations

Write

$$F(t,u) = 2\,\mathrm{arsinh}(\tan(t/2)\cos(u/2))\,, \qquad E_0(t,u) = F(t,u) - t,$$
$$Q_0(t,u) = \sqrt{1-\sin^2(t/2)\sin^2(u/2)}. \tag{4.1}$$

The subscript distinguishes these functions from the leaf coordinates E, Q. The square root is one at zero. Differentiation gives

$$F_t = \frac{\cos(u/2)}{\cos(t/2)Q_0}, \qquad F_u = -\frac{\sin(t/2)\sin(u/2)}{Q_0}. \tag{4.2}$$

Indeed $1+\tan^2(t/2)\cos^2(u/2) = Q_0^2/\cos^2(t/2)$, with matching branches at zero. The second derivative in (4.2) is symmetric in t, u, proving the integrability required by (1.3). The action is separately even and real on real arguments near zero.

Lemma 4.1. *On $|t|, |u| \le 10^{-2}$, both $|(E_0)_t|$ and $|(E_0)_u|$ are less than $\delta = 1/39000$. Moreover $E_0(t,u) = tH(t,u)$ for a holomorphic function satisfying $|H| \le \delta$, including at $t = 0$.*

Proof. Put $a = 1/79000$, $b = 1/199$, and $\omega = b^4$. With $r = 1/200$ the power-series estimates are

$$|\sin(t/2)| \le \frac{r}{1-r^2/6} < b, \qquad |\cos(t/2)-1| \le \frac{r^2}{2(1-r^2/12)} < a,$$

and likewise for u. Hence $|Q_0|^{-1} \le (1-\omega)^{-1}$ and $|Q_0 - 1| \le \omega$. For the latter use $(Q_0-1)(Q_0+1) = -\sin^2(t/2)\sin^2(u/2)$ and $\operatorname{Re} Q_0 > 0$. Equation (4.2) implies the rational bounds

$$|(E_0)_t| \le \frac{2a/(1-a)+\omega}{1-\omega} < \delta, \qquad |(E_0)_u| \le \frac{b^2}{1-\omega} < \delta. \tag{4.3}$$

Since $E_0(0,u) = 0$, take $H(t,u) = \int_0^1 (E_0)_t(st,u)\,ds$. This definition is holomorphic at zero and avoids division by an unknown that might vanish. □

Multiplying (1.4) by -2π gives

$$(d_v - \mathrm{i}n_v)\tau_v + \sum_{w\sim v} E_0(\tau_v,\tau_w) = -2\pi\mathrm{i}c. \tag{4.4}$$

Define the fixed-point map by solving for the first term. On the closed polydisk $|\tau_v| \le \rho_0 = 10^{-2}$ it satisfies, for complex $|c| \le 1$,

$$|\mathcal{T}_{c,v}| \le \frac{2\pi}{630} + \delta\rho_0 < \rho_0, \qquad \sum_w |\partial_{\tau_w}\mathcal{T}_{c,v}| \le \frac{2d_v\delta}{L_v} \le 2\delta < 1. \tag{4.5}$$

The strict radius inequality already follows from $\pi < 22/7$. The contraction theorem gives a unique solution; iteration from zero converges uniformly in c. Each iterate is holomorphic, so the solution is holomorphic. The strict radius margin allows $|c| \le 1+\eta$ for some $\eta > 0$, proving the neighborhood statement in Theorem 1.1.

Its sign is an additional fact, not a consequence of smallness. Equation (4.4) is

$$A_v\tau_v = -2\pi\mathrm{i}c, \qquad A_v = d_v - \mathrm{i}n_v + \sum_{w\sim v} H(\tau_v,\tau_w). \tag{4.6}$$

For real $c > 0$,

$$\operatorname{Re} A_v \ge (1-\delta)d_v > 0, \qquad \operatorname{Im}\tau_v = -\frac{2\pi c\,\operatorname{Re} A_v}{|A_v|^2} < 0, \qquad |A_v| \ge L_v - \delta d_v. \tag{4.7}$$

This proves (1.5). It divides by neither n_v nor τ_v, and works with mixed signs and zero powers. Substituting (1.2) and (1.4) in the peripheral formulas now gives $U_v + p_vV_v = -2\pi\mathrm{i}c$ as an equality of chosen logs.

5 The metric and the full primitive periods

Realize each edge block by Lemma 3.1, and glue its pages in the order of Proposition 2.1. Table (3.9) matches full complex shears, not just dihedral angles. Equations (3.11) and (3.12) check the disk edge returns. At every edge the angle branch is 2π, by positivity and continuation from the regular cells. Thus finite points have nonsingular hyperbolic neighborhoods. If an auxiliary subdivision uses a finite vertex, its topological sphere link has a nonsingular spherical metric; its complete spherical development is the ordinary sphere. No global injectivity of the developing map is needed here.

The horospherical triangles give a nonsingular similarity structure on each topological cusp torus. At a main corner the small meridian is $W_{j-1}^{-1}W_j$, or its inverse in the transported frame. Its multiplier is one. It remains nontrivial: for example $W_0^{-1}W_1$ is translation by $E-1$, and the other corner words are its marked counterparts. The small chart excludes $E = 1$. The commuting peripheral maps therefore are translations at this parabolic flag. The compact link is a Euclidean torus, so the main cusps are complete.

At an auxiliary end and real $c > 0$,

$$\operatorname{Re} V_v = -2 \operatorname{Im} \tau_v > 0. \tag{5.1}$$

The peripheral group has a nonunit dilation and hence a second common fixed point q_v in the affine chart whose first flag is infinity. Its similarity development misses q_v. Indeed the invariant vector field $(z - q_v)\partial_z$ descends to the torus. Any zero has index $+1$, because the developing map is locally orientation-preserving. The Euler characteristic of the torus is zero, so there are no zeros.

On the simply connected link cover define

$$f_v = \log(\mathrm{dev}_v - q_v). \tag{5.2}$$

Its real derivative is invertible. The metric $|df_v|^2$ descends to a nonsingular Euclidean metric on the compact torus and is complete. Lifting geodesics makes f_v a covering of the Euclidean plane, and therefore a diffeomorphism. Its two generator periods form an injective real rank-two lattice.

5.1 Fixing the winding integer

Let P_{μ_v}, P_{λ_v} be the periods of (5.2) along the original primitive paths. Their exponentials are the multipliers already computed, so

$$P_{\mu_v} - U_v, \;\; P_{\lambda_v} - V_v \in 2\pi \mathrm{i}\mathbb{Z}. \tag{5.3}$$

The integers are continuous for $0 < c \leq 1$: the link is nonsingular, the paths and flags are retained, and a base logarithm can be continued. Rebasing changes no period.

To determine the integers, use a continuous frame sending the chosen flag to infinity. The longitude is $z \mapsto a(c)z + b(c)$ with $a = e^{V_v}$, $a(0) = 1$, and $b(0) \neq 0$. The last inequality follows from the original primitive longitude in the complete Euclidean torus. Its second fixed point is $q_v = b/(1 - a)$; hence $|q_v| \to \infty$ as $c \downarrow 0$. The fixed-generator paths use finitely many triangles and continuous transition maps, so their images are bounded by some B and their Euclidean lengths by some L for this fixed tree. Thus

$$|P_\gamma(c)| \leq \frac{L}{|q_v(c)| - B} \longrightarrow 0. \tag{5.4}$$

Also $U_v, V_v \to 0$. The integers in (5.3) vanish near zero and hence along the whole path. In particular the actual primitive filling slope has period $-2\pi \mathrm{i}c$ even if the sum of its many small page contributions is close to $-2\pi\mathrm{i}$. No principal logarithm is reset.

5.2 Adding the core

Send the fixed points to zero and infinity. In upper half-space write $r = y/|z|$, and use coordinates

$$R_* = \operatorname{arsinh}(1/r), \qquad \theta = \operatorname{Im} f_v, \qquad X = \operatorname{Re} f_v + \tfrac{1}{2}\log(1 + r^2). \tag{5.5}$$

A direct substitution into $y^{-2}(|dz|^2 + dy^2)$ gives

$$dR_*^2 + \cosh^2 R_* \, dX^2 + \sinh^2 R_* \, d\theta^2. \tag{5.6}$$

The original ideal end contains all sufficiently high vertical rays over its finitely many similarity triangles. Compactness of their quotient gives a whole punctured tube $0 < R_* < R_0$ in these

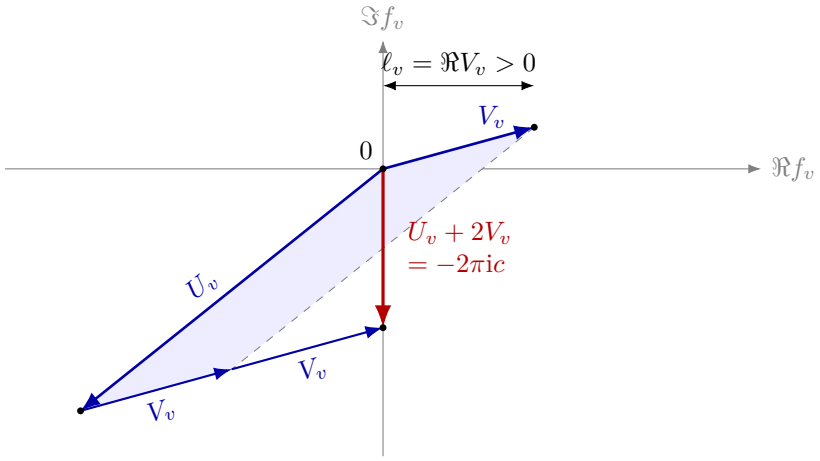


Figure 2: The auxiliary log-link plane for $c > 0$. The lattice generators are the full periods V_v and U_v. Their primitive combination $U_v + p_v V_v = -2\pi \mathrm{i} c$ is the angular meridian; V_v supplies the positive core translation. The diagram shows $p_v = 2$ for the period relation only. The real period of the slope vanishes, and its angular period reaches exactly one turn at $c = 1$.

coordinates. The primitive slope $\mu_v + p_v \lambda_v$ has zero real period and angular period $-2\pi c$. Its complementary curve is λ_v, because the change of primitive basis has determinant one. That curve has positive real period $\ell_v = \operatorname{Re} V_v$.

Adding $R_* = 0$ gives a cone solid torus of angle $2\pi c$ and core length ℓ_v. At $c = 1$ it is a smooth solid torus with precisely the original filling meridian. Main cusps are still complete horoball quotients and the middle is compact. Hence the resulting smooth metric is complete and has finite volume on $Y(\mathbf{p})$. Its universal local development covers simply connected hyperbolic space and is an isometry; this is the geometric holonomy of the original filled exterior. For $c < 1$ we assert a cone metric, not a smooth filled-group representation or discreteness of its holonomy.

6 Absolute volume and core estimates

We state precisely the local companion fact used in this section. For an oriented positive ideal triangulation of a torus-ended 3-manifold, suppose the full face and edge gluing equations hold and U_i, V_i are continued affine similarity logarithms along primitive oriented torus bases. The sum $\mathcal{V}_\rho$ of the oriented tetrahedral volumes satisfies

$$d\mathcal{V}_\rho = \frac{1}{2} \sum_i (\operatorname{Re} V_i \, d \operatorname{Im} U_i - \operatorname{Re} U_i \, d \operatorname{Im} V_i). \tag{6.1}$$

This is the local identity proved in [1, Section 7], equation (7.3); it does not assume that the ends are complete or that the total logs are small. Its convention is the positively ordered model and cross ratio (3.6). For one tetrahedron its proof starts from the Bloch–Wigner differential

$$dD(w) = \log|w| \, d \arg(1 - w) - \log|1 - w| \, d \arg w.$$

On the cusp triangles, the side-scale transitions telescope along the actual normal paths. The peripheral coefficient rows pair to minus twice their oriented intersection; after quotienting the isotropic edge rows the tangent pairing is $-\frac{1}{2} \sum_i dU_i \wedge dV_i$. These two identities give (6.1) for similarity logs. They are local chain identities and continue along a positive family. The classical framework appears in Neumann–Zagier [4].

The orientation of the primitive basis can also be checked directly in its exact complete coordinates. In the affine chart $h(z) = 1/(z - Z)$, with $E = \mathrm{i}$, $R = 1/2$, and $Z = (1 + \mathrm{i})/2$, the words S^{-1} and $W_2^{-1}W_0$ translate by $1 - \mathrm{i}$ and $2 + 2\mathrm{i}$. The negative and positive odd meridians translate by 2 and $-2\mathrm{i}$. Consequently the d_v-page collar has translations

$$\mu_v : \ d_v(1 - \mathrm{i}) - \epsilon_v(1 + \mathrm{i}), \qquad \lambda_v : \ 2(1 + \mathrm{i}), \qquad \operatorname{Im} \frac{\lambda_v}{\mu_v} = \frac{2d_v}{d_v^2 + \epsilon_v^2} > 0.$$

This fixes the same oriented primitive-basis convention used in (6.1). Here the quotient denotes the ratio of the displayed translation vectors, not a quotient of curves.

Here all hypotheses have been verified: Proposition 2.1 identifies the oriented manifold and primitive paths; Lemma 3.1 gives positive oriented tetrahedra; (3.9)–(3.12) supply complete edge returns; and Section 5.1 fixes the logarithmic lifts. Main meridians have $U = 0$ and contribute zero. Thus the companion's local differential applies to this whole path, without using its separate long-filling existence theorem.

Put $U_v^0 = -2\pi\Psi_{\tau_v}$ and define

$$\begin{aligned} G^0 &= 2\pi\mathrm{i}\left(\Psi - \sum_v \tau_v \Psi_{\tau_v}\right), \qquad dG^0 = \tfrac{1}{2}\sum_v V_v \, dU_v^0, \\ G &= G^0 - \sum_v \frac{\epsilon_v V_v^2}{8} = 2\pi\mathrm{i}\left(\Psi - \sum_v \tau_v \Psi_{\tau_v}\right) + \sum_v \frac{\epsilon_v \tau_v^2}{2}. \end{aligned} \tag{6.2}$$

Thus $dG = \frac{1}{2}\sum_v V_v dU_v$. Differentiating and using (6.1) shows that

$$\mathcal{V}_\rho = \operatorname{Im} G - \tfrac{1}{2}\sum_v \operatorname{Re} U_v \ \operatorname{Im} V_v. \tag{6.3}$$

The constant is fixed at the complete assembly: its $2e$ regular octahedra have volume $2ev_{\mathrm{oct}} = 2\pi\Psi(0) = \operatorname{Im} G(0)$. Every shape remains in a simply connected positive neighborhood of i, so no volume or logarithmic branch is reset.

6.1 The same fundamental chain computes metric volume

Equality (6.3) is not by itself an identification with filled metric volume. We make that comparison on the same eight-tetrahedron chain at each original tree edge. At an auxiliary end send its fixed points to zero and infinity, and retain one finite boundary triangulation. Let w range over a fixed fundamental cross section and let $Y \to \infty$ be a height parameter. The ideal conical map has the form $(a(w), Yb(w))$, where $b > 0$. The logarithmic link supplies a real function η whose deck increments are the real peripheral periods. The axis-valued map $(0, Ye^{\eta(w)})$ is equivariant. Compare them by

$$H_s(w, Y) = \left(sa(w), Y \exp\big((1 - s)\eta(w) + s \log b(w)\big)\right), \qquad 0 \le s \le 1. \tag{6.4}$$

The assignments agree on shared boundary faces. They are equivariant because a, b, and e^η transform by the complex multiplier, its modulus, and its modulus, respectively. On the compact cross section the horizontal derivatives are bounded, the height is comparable to Y, and its derivatives are $O(Y)$. The hyperbolic volume form $y^{-3}dx_1 \wedge dx_2 \wedge dy$ therefore bounds the boundary homotopy volume by $O(Y^{-2})$.

For fixed $0 < c \le 1$ this compares the ideal chain with the metric away from the cone axis; the axis has zero three-dimensional volume. At $c = 1$ the actual primitive meridian has period $-2\pi\mathrm{i}$ and zero real period, so the axis assignment extends over its meridian disk. Extend the same boundary triangulation through the solid torus with added vertices on the

axis. All added tetrahedra have zero volume. The main horoball comparisons likewise have vanishing boundary volume. Stokes' theorem on the common truncated chain and the limit in (6.4) give

$$\mathcal{V}_\rho = \mathrm{Vol}_c\,. \tag{6.5}$$

For $c < 1$ this is a comparison on the cone metric away from its axis, not an assertion of smooth group descent. At $c = 0$ it follows by continuity of the positive tetrahedral volumes.

At the critical point let $\Phi = G/(2\pi\mathrm{i})$. The equations (1.4) and $n_v = \epsilon_v - 2p_v$ give

$$\Phi - \mathcal{C}(c) = -\frac{\mathrm{i}}{2\pi}\sum_v p_v\tau_v^2, \qquad \tfrac{1}{2}\,\mathrm{Re}\,U_v\ \mathrm{Im}\,V_v = p_v\,\mathrm{Im}(\tau_v^2). \tag{6.6}$$

Substituting into (6.3) and (6.5) cancels the boundary correction and proves (1.8). This completes Theorem 1.1.

Corollary 6.1. *Let $S_{\mathcal{T}} = \sum_v d_v/L_v^2$. Under Theorem 1.1,*

$$\frac{4\pi c(1-\delta)d_v}{(L_v+\delta d_v)^2} \le \ell_v(c) \le \frac{4\pi c(1+\delta)d_v}{(L_v-\delta d_v)^2}, \tag{6.7}$$

and

$$\frac{2\pi^2(1-\delta)}{(1+\delta)^2}S_{\mathcal{T}} \le 2ev_{\mathrm{oct}} - \mathrm{Vol}(Y(\mathbf{p})) \le \frac{2\pi^2(1+\delta)}{(1-\delta)^2}S_{\mathcal{T}}. \tag{6.8}$$

Proof. In (4.6), $(1-\delta)d_v \le \mathrm{Re}\,A_v \le (1+\delta)d_v$ and $L_v - \delta d_v \le |A_v| \le L_v + \delta d_v$. Equation (4.7) gives (6.7). Stationarity in (1.7) gives

$$\frac{d}{dc}\mathrm{Vol}_c = 2\pi\sum_v \mathrm{Im}\,\tau_v(c) = -\pi\sum_v \ell_v(c).$$

Integrate from 0 to 1, and use $d_v \le L_v$ in the denominators. This proves (6.8). The derivative also agrees with the Schläfli expression $-\frac{1}{2}\sum_v \ell_v\, d(2\pi c)$; it is a normalization check, not the argument fixing the absolute constant. □

7 A criterion with different vertex radii

The common threshold is convenient but is not required by the proof. The next finite criterion uses the actual neighboring radii.

Theorem 7.1. *Choose $0 < r_v \le 10^{-2}$ and define*

$$s_v = \sinh(r_v/2), \quad a_v = \cosh(r_v/2) - 1, \quad \omega_{vw} = s_v^2 s_w^2, \tag{7.1}$$

$$\alpha_{vw} = \frac{a_v + a_w}{(1-a_v)\sqrt{1-\omega_{vw}}} + \frac{1}{\sqrt{1-\omega_{vw}}} - 1, \qquad \beta_{vw} = \frac{s_v s_w}{\sqrt{1-\omega_{vw}}}, \qquad h_v = \sum_{w\sim v}\alpha_{vw}. \tag{7.2}$$

Suppose

$$\begin{aligned} &2\pi + h_v r_v < L_v r_v, \qquad h_v < d_v, \\ &\max_v\left(\frac{h_v}{L_v} + \sum_{w\sim v}\frac{\beta_{vw}r_w}{L_v r_v}\right) < 1. \end{aligned} \tag{7.3}$$

Then the existence, cone completion, smooth endpoint, and absolute volume conclusions of Theorem 1.1 hold, with uniqueness in $|\tau_v| \le r_v$ and the bound

$$|\tau_v(c)| \le \frac{2\pi c}{L_v - h_v}, \qquad \mathrm{Im}\,\tau_v(c) < 0. \tag{7.4}$$

The last line of (7.3) may instead be replaced by a positive vector q and $\kappa < 1$ satisfying $Mq \le \kappa q$, where $M_{vv} = h_v/L_v$ and $M_{vw} = \beta_{vw}/L_v$ for adjacent distinct vertices. The invariant-polydisk inequalities in the first line remain required separately.

Proof. On the corresponding bidisk, (4.2) bounds $|(E_0)_t|$ by α_{vw} and $|(E_0)_u|$ by β_{vw}: use $|\sin(t/2)| \le s_v$, $|\cos(t/2) - 1| \le a_v$, and $|Q_0| \ge \sqrt{1-\omega_{vw}}$. The binomial series also gives $|Q_0^{-1} - 1| \le (1-\omega_{vw})^{-1/2} - 1$, which accounts for the final two terms in α_{vw}. Integrating $(E_0)_t$ from zero gives $|E_0(t,u)| \le \alpha_{vw}|t|$. The first inequality makes the fixed-point map preserve the product of the chosen disks. The last is its derivative bound in the norm $\max_v |\tau_v|/r_v$; alternatively $Mq \le \kappa q$ gives the q-weighted norm. Thus the same uniform iteration proves existence, uniqueness, and holomorphic dependence near $|c| \le 1$. The denominator now has real part at least $d_v - h_v > 0$ and modulus at least $L_v - h_v$. This proves (7.4). All geometric steps require only these sign and radius bounds, so Sections 5–6 apply unchanged. □

The inequalities are an explicit certificate in the integer degrees, powers, and chosen positive radii. Rational Taylor enclosures can verify them; they assume neither a saddle nor a representation. The contraction calculation itself works for a finite graph, but the geometric conclusion here uses the ordered tree cut proved in Proposition 2.1. It supplies no geometric identification for an arbitrary graph with cycles.

8 Use in filling asymptotics and finite verification

This paper exports a geometric implication: any analysis selecting the solution of (1.4) in the stated positive polydisk may identify its real critical action by (1.8). It does not provide the analysis that selects that branch from an oscillatory invariant. In particular a tree's Newton polynomial kernel and its turning factors require their own exact adapter; they do not follow merely from the abstract finite filling norm.

For reference, the common sufficient conditions in the retained circle-tree estimates all enter this geometric theorem. If

$$\sum_v \frac{d_v}{d_v^2 + n_v^2} \le 2^{-25}, \quad \text{or} \quad \max_v \frac{d_v}{d_v^2 + n_v^2} \le 2^{-120},$$

then respectively $L_v \ge 2^{25/2} > 630$ or $L_v \ge 2^{60} > 630$, since $d_v \ge 1$. The directed estimate uses $D_v = (d_v - \mathrm{i}n_v)/(2\pi)$ and requires $2^{-14}|D_v| \ge 2$, so $L_v \ge 2\pi\, 2^{15} > 630$. These observations identify their geometric input; none replaces the different remainder and noncancellation arguments of those estimates.

Two existing exact programs check bounded algebra used above: `audit_two_octahedron_angular_edge_block.py` checks the rational pants ratios, odd returns, angular formulas, and positive chart bounds, while `audit_all_circle_vertexwise_geometry.py` checks the rational inequalities in Lemma 4.1 and (4.5). They are in the repository's `experiments` directory and accept an explicit output path. The paper-local driver records their source hashes and results. These calculations check finite identities and constants; they do not prove the topological identification, the contraction theorem, the link covering argument, or the volume comparison. Those are the written arguments above.

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